TRIBHI KATHURIA

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My work focuses on hierarchical learning for robots operating in human-centered environments. I study how robots can learn high-level behaviors from demonstrations and human feedback on top of reliable low-level skills, and validate these on real robotic systems, from social navigation to medical robotics.

EDUCATION

University of Michigan | Ph.D. in Robotics — Sep 2020 – Dec 2024

Lab	Computational Autonomy and Robotics Lab (CURLY), PI: Maani Ghaffari Jadidi Interaction and Collaboration Research Lab (ICRL), PI: Xi Jessie Yang
Research Focus	Preference learning for robots that work with people, through Inverse Reinforcement Learning, Learning from Demonstration, and Behavior Planning.

University of Michigan | M.S. in Robotics (GPA: 3.80/4.00) — Sep 2018 – May 2021

Coursework	Self-driving Cars, Motion Planning, Mobile Robotics (SLAM), Reinforcement Learning, Math for Robotics, Robotic Systems Lab, Nonlinear Systems and Control.

NSIT, University of Delhi | B.E. in Instrumentation and Control Engineering, India — Aug 2014 – Jun 2018

ACADEMIC APPOINTMENTS

Robotics Institute, University of Michigan | Lecturer III — Sep 2025 – Present

- Instructor for ROB 101 (Computational Linear Algebra, about 170 students per semester) and ROB 320 (Robot Operating Systems, about 30 students per semester).
- Introduced flipped-classroom-style labs for ROB 101, asking students to translate math to Julia code while talking through the problem sets.
- ROB 101 liaison for the Distributed Teaching Collaborative (DTC), remotely supporting Howard students taking the course.
- Building on this collaboration, co-developed a proposal to extend ROB 101 to support AI education in R2 schools, presented at the AWS-MLU symposium: Link
- Developed lecture content and updated lab materials for ROB 320 for Fall 2025, supporting a project-based curriculum in which students built a ROS stack on an MBot platform and simulated a Fetch robot with correct kinematics.
- Taught Publish-Subscribe paradigm, where students used concepts from TCP communication over sockets and multi-threading to run velocity commands on an Omni-drive robot platform (Mbot).

GRANTS AND AWARDS

Emergent Ventures Grant, Mercatus Center, George Mason University | PI, $10K — 2026-2027

- Awarded to develop a remote AI/robotics education and mentoring program connecting undergraduate and high school students in countries such as India with graduate student mentors, using a curriculum I developed.

Towner Prize for Outstanding Graduate Student Instructors | Honorable Mention — 2022

- University of Michigan. Award listing: https://crlte.engin.umich.edu/towner-prize-winners/

RESEARCH EXPERIENCE

University of Michigan | Independent Research — Sep 2025 – Present

- Developing **Hierarchical Reinforcement Learning** based navigation for dense, crowded environments, where a high-level policy switches between pretrained low-level skills using the robot's first-person and top-down images.
- Submitted a paper to ICRA 2027 presenting a hierarchical planner evaluated on a narrow crossing scenario.
- Beginning work on **preference learning** from da Vinci surgical camera data with Wayne State University.
- Mentor undergraduate students in creating datasets of socially aware scenarios in photo-realistic simulation.

University of Warwick and University of Leeds | Visiting Researcher, UK — Aug 2026

- Hosted by Prof. Emma MacPherson (University of Warwick) and Prof. Pietro Valdastri (University of Leeds); conducted robotic THz experiments at Warwick.
- Developed a **Gaussian process regression** model that estimates probe-skin contact quality from force-sensor feedback and the THz sensor posterior.
- Used the model to correct probe contact during robotic THz scanning, enabling reliable readings and reducing clinical scan time.
- Submitted a paper to ICRA 2027 on sparse contact prediction and probe correction for robotic THz scanning.
- Continuing the work remotely, now dynamically closing contact gaps by moving the robot with the scanning stage.

University of Michigan | Graduate Student Research Assistant — Sep 2020 – Dec 2025

- Developed motion and behavior planning algorithms for Socially Assistive Robots navigating dense, dynamic indoor environments.
- Developed an **Inverse Reinforcement Learning** based local planner to learn robot navigation behaviors from raw sensor data. Presented results on **crowd navigation** and **deadlock avoidance** for robots navigating around humans.
- Incorporated reasoning from **foundation models** to differentiate good vs. bad social behavior and supervise the ranking of demonstration data.
- Designed a **Mixed Integer Linear Programming** based tour planner that handles multiple visitor constraints to plan routes on a spatial-semantic graph, used as the global planner for an assistive robot.

Perceptual Robotics Lab (PeRL), University of Michigan | Research Assistant Sep 2019 – Sep 2020

- Implemented an MDP planner for navigating a museum with a Fetch robot, tested in simulation in Gazebo.
- Implemented **Augmented MDP** and **MC-POMDP** (approximate POMDP) planners on a simulated grid-world problem.

Honda R&D Americas | Automated Vehicle Control Prototyping Intern, Ann Arbor, MI May 2019 – Dec 2019

- Developed a control algorithm to carry out the longitudinal control of the vehicle for a custom application. Communicated with a DSRC roadside unit (V2I) to get information about the remote vehicles to the host vehicle for decision making and control.
- Controlled the speed and the brake of the vehicle, communicating messages to the Vehicle CAN via a MicroAutobox (RCP).
- Planned the trajectory of the vehicle for L2 autonomy using a pure pursuit algorithm with GPS data for feedback. Successfully completed real-world testing of the application for a company showcase.

PROJECTS

Robot Guide at the University of Michigan Museum of Art Jun 2022 – Aug 2024

- Used a **GPT davinci-002 model** agent to fine-tune responses from the robot to give specific answers about the artworks at the museum, including contextual prompts like "Tell me more about this?", with ROS support. Code available at: https://github.com/UMich-CURLY/Hi-Fetch
- Built an offline map of the museum and localized online to plan and complete a tour with small groups of participants walking behind the robot. The agent introduced the artwork and then could handle follow-up questions.
- Mentored an iOS app development project that let the robot showcase museum exhibits and introduce them to a visiting class of high school students.
- Demo video available at: https://youtu.be/SgwyzEO0flY?si=7JQ-e9seXoiLh7Ym

Motion Planning under Uncertainty Jan 2019 – Mar 2019

- Implemented discrete (A*, ANA*) and sampling-based (RRT, Bi-RRT) motion planners for the PR2 robot in a cluttered environment in OpenRAVE.
- Proposed a particle RRT where the different particles were used to characterize the uncertainty in the robot model and do re-planning when the maximum likelihood particle trajectory was far from the planned trajectory.

Robotic Systems Lab, University of Michigan Sep 2018 – Dec 2018

- Balancebot Project: Designed a PID-controller for the balancing motion of a two-wheeled robot and drove it autonomously in a maze captured by a motion capture system on a path planned using a Virtual Potential Field.
- Mobilebot Project: Performed Monte-Carlo localization for a LiDAR equipped mobile robot and used A* path planning for optimal exploration to map an unknown environment.

PUBLICATIONS

- X. Wang, **T. Kathuria**, **(Under Review)** Hierarchical Policy Learning for Socially Aware Robot Navigation. *2027 IEEE International Conference on Robotics and Automation (ICRA).*
- X. Guo, **T. Kathuria** et al., **(Under Review)** Sparse Contact Prediction and Probe Correction for Robotic Terahertz Scanning. *2027 IEEE International Conference on Robotics and Automation (ICRA).*
- **T. Kathuria** et al., **(2025)** Learning Implicit Social Navigation Behavior using Deep Inverse Reinforcement Learning. *IEEE Robotics and Automation Letters (RA-L).* IEEE Xplore
- Y. Xu, T. Chakhachiro, **T. Kathuria** et al., **(2023)** SoLo T-DIRL: Socially-Aware Dynamic Local Planner based on Trajectory-Ranked Deep Inverse Reinforcement Learning. *2023 IEEE International Conference on Robotics and Automation (ICRA).* DOI: 10.1109/ICRA48891.2023.10160536
- **T. Kathuria** et al., **(2022)** Providers-Clients-Robots: Framework for spatial-semantic planning for shared understanding in human-robot interaction. *31st IEEE International Conference on Robot & Human Interactive Communication (ROMAN).* DOI: 10.1109/RO-MAN53752.2022.9900525
- B. Fu, **T. Kathuria** et al., **(2021)** Simultaneous Human-Robot Matching and Routing for Multi-robot Tour Guiding under Time Uncertainty. *Journal of Autonomous Vehicles and Systems Special Issue.* DOI: 10.1115/1.4053428

- **T. Kathuria** et al., **(2018)** Control of Three Link Manipulator using Fractional Order PID Controller. *Fractional Order Systems, 2018 - Elsevier.* DOI: 10.1016/B978-0-12-816152-4.00016-9
- **T. Kathuria** et al., **(2017)** Study of Optimization Methods for Tuning of PID Gains for Three Link Manipulator. *Proc. of 7th International Conference on Cloud Computing, Data Science & Engineering - Confluence.* DOI: 10.1109/CONFLUENCE.2017.7943131

PRESENTATIONS

- ICRA 2023 Workshop L3H2 - Lifelong Learning with Human Help. Presented work title: Human-Aware motion planning using IRL on spatial semantic graphs. https://life-long-learning-with-human-help-l3h2.github.io
- ICRA 2022 Workshop ROPM - Robotic Perception And Mapping . https://sites.google.com/view/ropm/home
- ROMAN 2022 Talk - Providers-Clients-Robots: Framework for spatial-semantic planning for shared understanding in human-robot interaction. https://doi.org/10.48448/eh1p-8060
- COMCO 2021 - Computational Cognition Conference. Planning architecture for SARs (Poster). Archived abstract

PREVIOUS TEACHING EXPERIENCE

University of Michigan | Graduate Student Instructor, Self-Driving Cars — Sep 2023 – Dec 2023
- Held regular office hours, with emphasis on control theory.
- Gave a guest lecture on task planning and routing for self-driving vehicles, with hands-on examples in Python.

University of Michigan | Graduate Student Instructor, Computational Linear Algebra — Sep 2020 – Dec 2021
- Co-created content for the Pilot offering of a Linear Algebra course for Freshmen, including lab work and recitation material. Led the LiDAR project development.
- Led regular recitations. Videos and content: https://github.com/michiganrobotics/rob101

University of Michigan | Guest Lecturer, Mobile Robotics — Apr 2019
- Lecture on Motion Planning for Autonomous Systems. Introduction to Reinforcement learning, and background with MDP and POMDPs.

PROFESSIONAL SERVICE

Departmental Service
- Member, Undergraduate Curriculum Committee (UGCOM), University of Michigan Robotics — 2025–Present
- Member, Undergraduate Awards Committee, University of Michigan Robotics — 2025–Present
- Faculty Advisor, Robotics Undergraduate Student Committee (RUSC) — 2025–Present
- Qualifying Exam (CQE) Examiner, University of Michigan Robotics — 2025–2026
- Participated in faculty hiring discussions

Peer Reviewing
- IEEE Transactions on Robotics (T-RO) - 2024, 2025, 2026
- IEEE Transactions on Automation Science and Engineering (T-ASE) - 2026
- IEEE Robotics and Automation Letters (RA-L) - 2022, 2024, 2025, 2026
- IEEE International Conference on Robotics and Automation (ICRA) - 2020, 2022, 2023, 2024, 2025
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) - 2020, 2022, 2023, 2024, 2025, 2026
- International Conference on Ubiquitous Robots (UR) - 2020

Mentoring
- Xinyuan Wang (Hierarchical Policy Learning for Social Navigation) — Current: Undergraduate Student, UMich
- Xiangyu Guo (Robotic THz Scanning) — Current: Ph.D. Student, University of Warwick
- Yifan Xu (Global Task/Tour Planning) — Next Position: Ph.D. Student, UMich
- Theodor Chakhachiro (Global Tour/Task Planning) — Next Position: Controls Engineer, Figure AI
- Ke Liu (Navigating narrow environments) — Next Position: Software Engineer, ByteDance/TikTok
- Erik Chi (Tour Guide Demo) — Next Position: Ph.D. Student, UMich

Outreach
- Shared Robotics resources and experiences in GuruTalks: https://youtu.be/hKLp_sW0t9A — 2022
- Mentoring undergraduate DEI scholars — 2021, 2022
- Student Council Social Chair — 2021
- Member of Robotics DEI committee — 2021
- Mentor for UMich Robotics Mentorship Program — 2019

TECHNICAL SKILLS

- **Machine Learning Tools:** Pytorch, Tensorflow, Open AI Gym, CUDA, Tensorboard, Visdom.
- **Robotics and Simulation:** ROS, ROS Navigation Stack, Gazebo, lcm, OpenRAVE, CANalyzer, LABVIEW.
- **Programming Languages:** C++, C, Python, Julia, MATLAB/Simulink, ControlDesk.
- **Tools/Frameworks:** Linux, git, WSL, Docker, Jupyter, Conda.